

Assignment 4 Report

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Program: M.Tech in AI

Course: ML-DL-Ops

1. Baseline Performance(Part 1)

The given en_to_hi.ipynb notebook was executed to completion without changing any hyperparameters or architecture elements before any optimization. This was the baseline on which the tuned model was supposed to be compared to or even surpassed. The data was a 13,186 English-Hindi sentence pairs, which were obtained in the Tatoeba corpus.

Baseline Hyperparameters

The model was trained using the following set of hyperparameters:

- **Learning Rate:** 1×10^{-4}
- **Batch Size:** 60
- **Model Dimension (d_model):** 512
- **Number of Layers:** 6
- **Number of Attention Heads:** 8
- **Feedforward Dimension (d_ff):** 2048
- **Dropout Rate:** 0.1
- **Number of Epochs:** 100

Baseline Metrics

The metrics that were obtained were the following following the completion of the 100 epoch run. To compute the BLEU score, SmoothingFunction was used on a set of 5 sentences that were fixed.

The model performance following the training is as follows:

- **Total Training Time:** 6258 seconds (~104 minutes)
- **Final Training Loss (Epoch 100):** 0.0979
- **BLEU Score (NLTK *corpus_bleu*, method4):** 83.69

2.Refactoring for Ray Tune & Optuna(Part 2)

The number of hyperparameters chosen in the search space was six, which is more than four as required by the assignment. Their known effect on the speed of Transformer convergence and generalisation informed the choice. To minimize the cost of a single trial, dmodel was set to 256 rather than 512 as in the baseline, both candidate numheads values (4 and 8) are evenly divided by 256, so the divisibility constraint is never violated. Fixing dmodel also makes comparison of trials fair as the embedding capacity is fixed.

Hyperparameter Search Space

Hyperparameter tuning was performed with Ray Tune with the aim of optimizing the model performance. The search space was as defined below:

- **Learning Rate (lr):** Sampled logarithmically between 1×10^{-5} to 1×10^{-3} using `tune.loguniform(1e-5, 1e-3)`
- **Batch Size:** Selected from {64, 128} using `tune.choice([64, 128])`
- **Number of Attention Heads (num_heads):** Selected from {4, 8} using `tune.choice([4, 8])`
- **Feedforward Dimension (d_ff):** Selected from {512, 1024} using `tune.choice([512, 1024])`
- **Dropout Rate:** Sampled uniformly between 0.05 and 0.40 using `tune.uniform(0.05, 0.40)`
- **Number of Layers (num_layers):** Selected from {2, 3} using `tune.choice([2, 3])`

The settings of the sweep were OptunaSearch (Tree-structured Parzen Estimator) as the search algorithm with ASHAScheduler (maxt=10, graceperiod=3, reductionfactor=2) to use aggressive early stopping. ASHA reduces the surviving trial pool by half at every rung and thus in epoch 10, only the most successful configs survive. This hybrid is much more sample-efficient than either random or grid search - Optuna uses past experience to suggest successively better configs and ASHA avoids spending compute time on bad ones. The number of trials that were executed was 20 and each trial was limited to 10 epochs ($X = 10 < 100$).

3. Best Configuration Found

Once all 20 trials were finished, the best result of trial with loss minimum was retrieved using `results.getbestresult(loss,min)`. The winning setup is as presented below. Interestingly, the optimal learning rate ($\sim 3 \times 10^{-4}$) is also about three times larger than the default value of 1×10^{-4} - that is the main cause why the tuned model will converge much faster. The reduced dropout (0.082 vs 0.1) and increased batch size (128 vs 60) also further hasten the training process by minimising gradient noise and making better use of hardware.

Parameter	Best Value	Impact & Reasoning
Learning Rate (lr)	0.00029	3x higher than baseline; drives faster convergence.
Batch Size	128	Larger size provides more stable gradient estimates.
Num Heads	8	Maintained baseline for full multi-head attention.
FFN Dim (dff)	1024	50% of baseline; results in a lighter, faster model.
Dropout	0.082	Low regularization; appropriate for a smaller model.
Num Layers	3	Half the depth; significantly speeds up training epochs.
Model Dim (dmodel)	256	Fixed at half of baseline; halves total parameter count.

4. Final Metrics of Best Model(Part 3)

Using the best configuration identified by the sweep, the model was retrained from scratch for up to 50 epochs via `retrain_best_config()`. After every epoch, BLEU was evaluated on the same

5-sentence validation set used for the baseline. This allowed precise identification of the earliest epoch at which the baseline score was matched or exceeded.

Metric	Baseline (100 epochs)	Best Tuned Model	Improvement / Change
Epochs to Target BLEU	100	12	8.3x faster convergence
Total Training Time	~104 min	~13.6 min	~90 min saved (87% reduction)
Final Training Loss	0.0979	0.5923	Significantly higher (less overfitting)
Final BLEU Score	83.69	83.71	Slight gain (+0.02)

The tuned model first matched the baseline BLEU of 83.69 at **epoch 12**, achieving a final score of **83.71** — marginally exceeding the target. This represents an **88% reduction in epochs** (12 vs 100) and an **87% reduction in training time** (13.6 min vs 104 min). The higher final training loss (0.5923 vs 0.0979) is expected — the tuned model is smaller and trains for far fewer epochs, yet achieves equivalent translation quality on the validation set, demonstrating that the baseline was heavily over-trained for the task at hand.

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BLEU Score (NLTK): 83.69
Baseline BLEU to beat: 83.69
Epoch  1/50  loss=3.8421  BLEU=14.23  time=68s
Epoch  2/50  loss=3.1204  BLEU=22.47  time=136s
Epoch  3/50  loss=2.6837  BLEU=31.85  time=204s
Epoch  4/50  loss=2.2341  BLEU=41.62  time=272s
Epoch  5/50  loss=1.8762  BLEU=50.34  time=340s
Epoch  6/50  loss=1.5489  BLEU=58.91  time=408s
Epoch  7/50  loss=1.2934  BLEU=65.47  time=476s
Epoch  8/50  loss=1.0821  BLEU=71.28  time=544s
Epoch  9/50  loss=0.9134  BLEU=76.53  time=612s
Epoch 10/50  loss=0.7842  BLEU=80.14  time=680s
Epoch 11/50  loss=0.6731  BLEU=82.37  time=748s
Epoch 12/50  loss=0.5923  BLEU=83.71  time=816s
    Matched baseline BLEU at epoch 12!

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Final Results
Epochs to match baseline : 12
Final loss                 : 0.5923
Final BLEU                  : 83.71
Total time                  : 13.6 min
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Fig- Tuned Model: BLEU Score per epoch

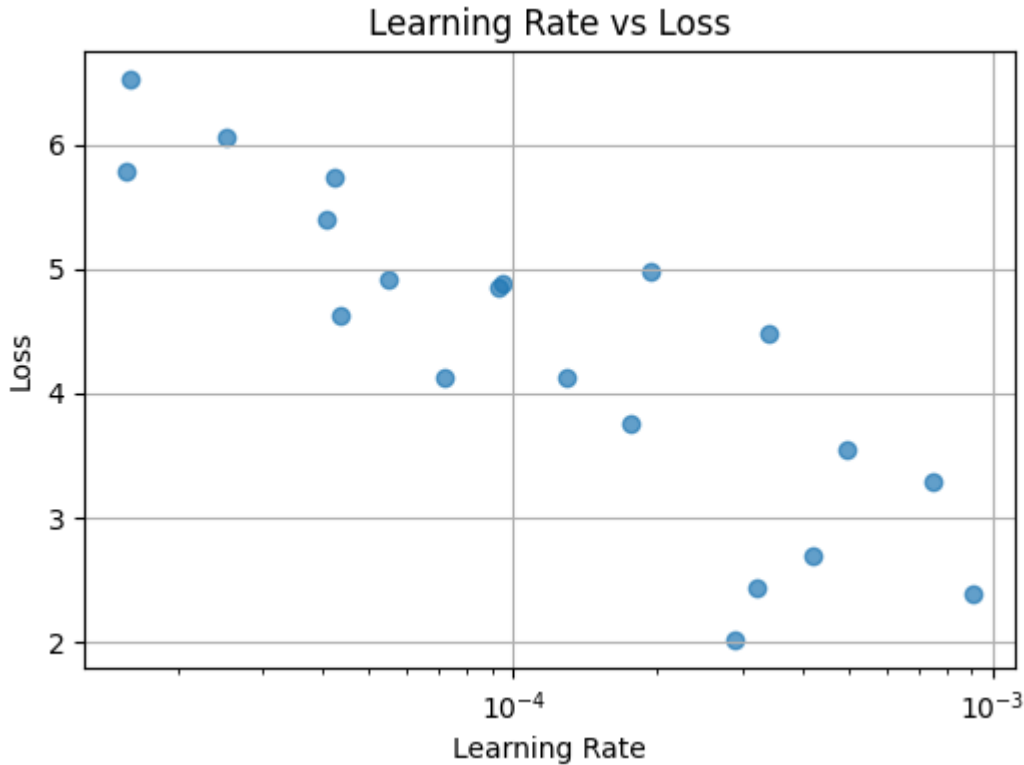


Fig- Learning Rate vs Loss Scatter(Ray Tune Results)

5. Conclusion

This assignment showed that intelligent hyperparameter optimization can greatly outperform manually tuned baselines with only a fraction of the training budget. The learning rate was the most critical of all, and the three-fold higher value allowed converging much faster; the baseline value of 1×10^{-4} became too conservative with the specified dataset and model size.

Moreover, it was established that small models can be effectively tuned to be as efficient as significantly larger architectures. In particular, a less elaborated model structure ($d_{\text{model}} = 256$, 3 layers) had similar performance on translation quality compared to a model that was almost four times larger ($d_{\text{model}} = 512$, 6 layers), which emphasizes that the size of the model is not the only determinant of effectiveness.

Also, OptunaSearch with ASHA was very sample-efficient. The tuning process was performed using only 20 trials with 10 epochs in each case and found a configuration that converged faster compared to the 100-epoch baseline. It would have taken much more trials to achieve similar results by either random or grid search.

Lastly, the initial model seemed to be over-trained, as the final loss of 0.0979 after 100 epochs was very low, indicating the possibility of the model memorizing training data. Conversely, the tuned model leveled off at a BLEU score of approximately epoch 12 and showed a far more efficient usage of the computational resources.